1. **B.Tech, COURSES OFFERED BY DEPARTMENT OF MATHEMATICS from 5th to 8th semester.**

Course 1: Financial Engineering

A. OVERVIEW OF THE COURSE:

Name of the Course	**Financial Engineering**
Offering Department	**Mathematics**
B.Tech Branches to which this course is offered	**Mathematics and Computing (MAC)**

Overview:

- The course financial Engineering is offered in the **fifth semester** to the Mathematics and Computing (MAC), a branch of B. Tech (CSE). The objective is to provide fundamental and essential mathematical tools to study the problems arising in the area of finance. Apart from consisting two Nobel Prizes winning theories of Black, Scholes and Merton for option pricing and Mean-Variance approach of Markowitz for portfolio optimization, the course consists of topics of understanding of the financial markets, value of money, bond pricing and risk management. At the end of this course, students will be able to calculate the theoretical prices of bonds/option derivatives and also competent of optimizing the real time investment & managing its risk.

B. Syllabus

Course No	Title of the Course	Course Structure	Prerequisite
---------	Financial Engineering	3L-1T-0P	None

COURSE OUTCOMES (COs): After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

CO1: Know the basics of financial engineering, financial markets, and financial instruments.
CO2: Understand the concepts of interest rates, value of money and working of bonds
CO3: Understand the option pricing in the framework of discrete and continuous time
CO4: Formulate and solve mean-variance portfolio optimization model.
CO5: Construction of portfolio optimization models with other risk measures and calculation of tail risk measures.

COURSE CONTENT

UNIT I: Introduction to financial instruments, interest rates and pricing of bonds: Introduction of financial system, financial markets, and financial instruments: stocks, bonds, derivatives, mutual funds; Interest rates (nominal, real, and effective), time value of money, present and future values of cash flow streams; Bonds and bonds pricing, yield, duration, term structure of interest rates, forward rates.

UNIT II: Option pricing I: Introduction to derivatives, concept of hedging, no-arbitrage principle, forward and futures contracts and their pricing with and without dividends, hedging strategies using futures; Call and put options, hedging strategies involving options, Pay-off curves of options combinations, single and multi period binomial lattice models for European and American options pricings, existence of risk neutral probabilities, options on dividend paying stock.

UNIT III: Option pricing II: The Cox-Ross-Rubinstein (CRR) formula, Black-Scholes option pricing formula.

UNIT IV: Portfolio optimization I: Expected value and variance of a random variable, concept of diversification, mean-variance portfolio optimization (Markowitz model), solving the Markowitz model using real life stock price data, Capital asset pricing model (CAPM).

UNIT V: Portfolio optimization II: Mean absolute deviation (MAD) based portfolio optimization, Minimax rule based portfolio optimization, Introduction and calculation of Value-at-Risk (VaR) of an asset and Conditional Value-at-Risk (CVaR) of an asset.

SUGGESTED READINGS

Text book:

1. S. Chandra, S. Dharmaraja, A. Mehra, R. Khemchandani , Financial Mathematics: An Introduction, Alpha Science International Ltd.

SUGGESTED READINGS:

1. M. Capinski and T. Zastawniak (2003), Mathematics for Finance: An Introduction to Financial Engineering and Springer-Verlar, London
2. D. G. Luenberger (1998), Investment Science, Oxford University Press, New York.
3. J. C. Hull (2000), Options, Futures and other Derivatives, Fourth edition, Prentice Hall Inc., Upper Saddle River.
4. R. J. Elliott and P. E. Kopp, Mathematics of Financial Markets, Springer, 1999.
5. M. Bartholomew-Biggs, Nonlinear optimization with financial applications, Springer Science & Business Media, 2005.

C. . CO-PO & CO-PSO MAPPING TABLE

CO\ PO	P O1	P O2	P O3	P O4	P O5	P O6	P O7	P O8	P O9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	1	2	1	0	0	0	0	0	0	0	0	0	3	1	1
CO2	3	3	3	0	2	0	0	0	2	2	1	1	3	3	0
CO3	2	2	3	0	1	0	0	0	2	2	1	1	2	2	0
CO4	1	2	3	0	0	0	0	0	1	1	1	1	2	2	0
CO5	1	1	2	0	0	0	0	0	1	1	1	1	2	2	0

D. LESSON PLAN for THEORY LECTURE

Lecture No.	Topic to be covered	No. of lectures	Unit
1	Introductory lecture on financial system, financial markets and financial instruments	1	Unit 1 (10)
2	Types of interest rates (nominal, real, and effective), time value of money.	1	
3	Calculation of present and future values of cash flow streams.	2	
4	Calculation of bonds pricing	2	
6	Relationship between price of bond, time to maturity, quality ratings, yield.	2	
7	Bond duration	1	

8	Term structure of interest rates, forward rates.	1	
9	Introduction to derivatives, no-arbitrage principle	1	Unit 2 (11)
10	Forward and futures contracts and their pricing	3	
11	Introduction to call and put options, hedging strategies involving options	1	
12	Pay-off curves of options combinations,	1	
13	Single and multi period binomial lattice models for European and American options pricings	4	
15	Existence of risk neutral probabilities and pricing of options on dividend paying stock.	1	
	MID SEMESTER EXAMS		
26	The Cox-Ross-Rubinstein (CRR) formula: statement, proof and solving examples	3	Unit 3 (7)
27	Black-Scholes option pricing formula: statement, proof and solving examples	3	
28	Black-Scholes option pricing formula for dividend paying stock: solving examples	1	
29	Introduction to portfolio optimization and Expected value and variance of a random variable	2	Unit 4 (7)
30	Mean-variance portfolio optimization (Markowitz model): concept of diversification, solving the Markowitz model using Lagrange method, Markowitz bullet.	4	
31	Capital asset pricing model (CAPM).	1	

34	Mean absolute deviation (MAD) based portfolio optimization: Introduction and formulation	1	Unit 5 (6)
35	Minimax rule based portfolio optimization: Introduction and formulation	1	
36	Tail risk management: Introduction and calculation of Value-at-Risk (VaR) of an asset	2	
37	Tail risk management: Introduction and calculation of Conditional Value-at-Risk (CVaR) of an asset.	2	

MID SEMESTER EVALUATION	
	Based on the units the evaluation will be done
	CLASS TEST-1
	As per the schedule decided in academic calendar
	CLASS TEST-2
	As per the schedule decided in academic calendar
END SEMESTER EVALUATION	

F. SELF STUDY

Sr. No.	**Topic**	
1	To prepare notes from the different books.	Unit I-V
2	To solve practice problems from reference books etc.	